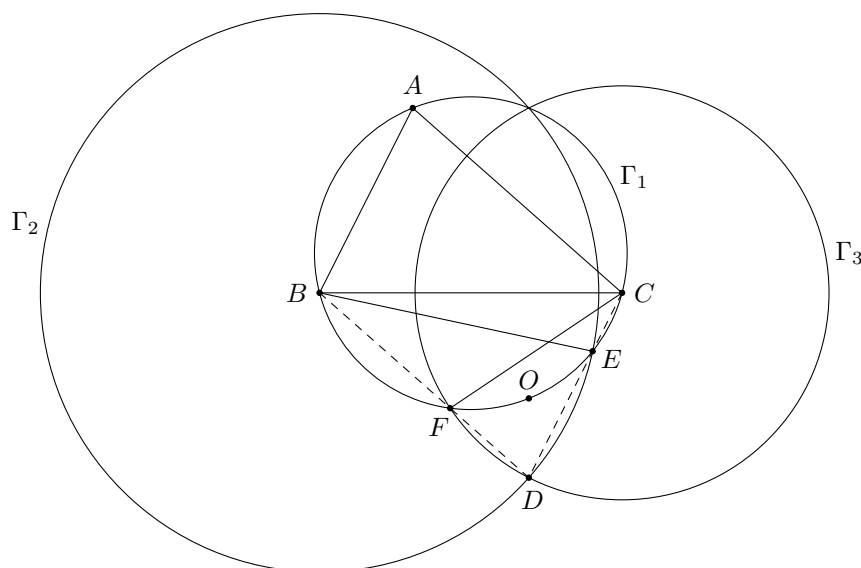




HK IMO Training 2023-2024 Problem Set 17 Suggested Solutions



Problem 65 (Greece TST 2022 Problem 2). In $\triangle ABC$, $AB < AC < BC$. Let Γ_1 be the circumcircle of $\triangle ABC$, let Γ_2 be the circle with centre B and radius AC , let Γ_3 be the circle with centre C and radius AB . Suppose D is an intersection point of Γ_2 and Γ_3 , E is an intersection point of Γ_1 and Γ_2 , F is an intersection point of Γ_1 and Γ_3 , such that D, E, F lie on the opposite side of BC as A . Prove that the circumcentre of $\triangle DEF$ lies on Γ_1 .



Solution. Since $BD = AC$ and $CD = AB$, $ABDC$ is a parallelogram. This implies $AC \parallel BD$. Also, as $CF = AB$ and $ABFC$ is cyclic, $ABFC$ is an isosceles trapezoid. This implies $AC \parallel BF$. Thus, B, F, D are collinear. Similarly, C, E, D are collinear.

Let O be the circumcentre of $\triangle DEF$. Then we have

$$\angle EOF = 2\angle EDF = 2\angle CDB = 2A$$

and

$$\angle EBF = \angle CBF - \angle CBE = C - (\angle ABE - B) = C - A + B$$

since $ABEC$ is an isosceles trapezoid. It follows that

$$\angle EOF + \angle EBF = 2A + (C - A + B) = A + B + C = 180^\circ.$$

Therefore, B, F, O, E are concyclic. This means O lies on Γ_1 . □

Problem 66 (Francophone MO 2022 Senior Problem 4 modified). Find all positive integers m and n such that all digits of $(2m)^n + 1$ in the decimal representation are equal.

Solution. Answer: $(m, n) = (1, 2), (1, 3), (1, 5), (3, 5), \left(\frac{1}{2} \left(\frac{a(10^k - 1)}{9} - 1 \right), 1\right)$ where $a = 1, 3, 5, 7, 9$ and $k \geq 1$ except $a = k = 1$.

When $n = 1$, all digits of $2m + 1$ are the same. Since $2m + 1$ is odd, we must have $2m + 1 = \frac{a(10^k - 1)}{9}$ for some $a = 1, 3, 5, 7, 9$ and $k \geq 1$. As $m \geq 1$, the case $a = k = 1$ is rejected. This gives the last class of solutions.

Suppose $n \geq 2$. Clearly, the condition is satisfied if $(2m)^n + 1$ is a one-digit number. As $9 \geq (2m)^n + 1 \geq (2m)^2 + 1$, we have $m = 1$. In that case, only $n = 2, 3$ satisfy $2^n + 1 \leq 9$. Thus, two solutions are $(m, n) = (1, 2), (1, 3)$. Now, suppose $(2m)^n + 1$ has at least two digits.

- If all digits of $(2m)^n + 1$ are 1, 5 or 9, then $(2m)^n + 1 \equiv 11, 55, 99 \equiv 3 \pmod{4}$. However, we also have $(2m)^n + 1 \equiv 1 \pmod{4}$ since $4 \mid (2m)^n$ for $n \geq 2$. This is a contradiction.
- If all digits of $(2m)^n + 1$ are 3, either $(2m)^n + 1 = 33$ or it has at least three digits. In the former case, we solve $(2m)^n = 32$ to get a unique solution $(m, n) = (1, 5)$ for $n \geq 2$. In the latter case, we have $(2m)^n + 1 \equiv 333 \equiv 5 \pmod{8}$. This implies $n = 2$ (or otherwise $(2m)^n + 1 \equiv 1 \pmod{8}$). But then $(2m)^2 \equiv 3 - 1 = 2 \pmod{5}$ has no solution.
- If all digits of $(2m)^n + 1$ are 7, either $(2m)^n + 1 = 77, 777, 7777$ or it has at least five digits. Note that $77 - 1 = 76$ and $777 - 1 = 776$ are not perfect powers, while $7776 = 6^5$. This yields a solution $(m, n) = (3, 5)$. In the last case, we have $(2m)^n + 1 \equiv 77777 \equiv 17 \pmod{32}$. This implies $v_2((2m)^n) = 4$, and hence $n = 2, 4$. But then $(2m)^n \equiv 0 - 1 \equiv 6 \pmod{7}$ has no solution as 6 is not a quadratic residue modulo 7.

Therefore, we have $(m, n) = (1, 2), (1, 3), (1, 5), (3, 5)$ for $n \geq 2$. □

Problem 67 (All-Russian Olympiad 2015 Grade 11 Problem 3). There are 110 participants in a tournament. Every pair of participants plays against each other exactly once, and there is no draw in each game. After all the games are played, it is found that for any 55 participants, there is one of them who has lost to at most 4 of the other 54 participants. Prove that there is a participant who has lost to at most 4 of the other 109 participants in the tournament.

Solution. We prove the following statement $P(n)$ for any positive integer $n = 55, 56, \dots, 110$: For any n participants, there is one of them who has lost to at most 4 of the other $n - 1$ participants.

The base case $n = 55$ is the given condition. Assume $P(n)$ is true for some $n = 55, 56, \dots, 109$. Consider any set S of $n + 1$ participants $A_1, A_2, \dots, A_{n+1}$. Suppose on the contrary that none of them has lost to at most 4 of the other n participants. For each $i \in \{1, 2, \dots, n + 1\}$, if we remove A_i , there is a participant $A_{f(i)}$ with $f(i) \neq i$ who has lost to at most 4 of the remaining participants. By our assumption, $A_{f(i)}$ must lose to exactly 5 participants in S including A_i .

Consider the set $T = \{A_{f(i)} : i = 1, 2, \dots, n + 1\}$, and let $t = |T|$. Since each participant in T has lost to exactly 5 participants in S , each of them corresponds to at most 5 different $f(i)$'s. Therefore, we have

$$t \geq \left\lceil \frac{n+1}{5} \right\rceil \geq \left\lceil \frac{56}{5} \right\rceil = 12.$$

We focus on the t participants in T . They have played $\binom{t}{2} = \frac{t(t-1)}{2}$ games in total. By the pigeonhole principle, there exists one of them who has lost to at least $\frac{t-1}{2} > 5$ participants in T . This contradicts the fact that every participant in T has lost to exactly 5 participants in S (hence at most 5 participants in T). Therefore, $P(n + 1)$ is true.

By induction, $P(n)$ is true for all $n = 55, 56, \dots, 110$. The final result is exactly $P(110)$. □

Problem 68 (RMM Shortlist 2021 A4). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a monotonic increasing function such that

$$f(y) - f(x) < y - x \tag{1}$$

for all $y > x$. Consider any sequence $a_1, a_2, \dots$ of real numbers satisfying

$$a_{n+2} = f(a_{n+1}) - f(a_n)$$

for all $n \geq 1$. Prove that for any $\varepsilon > 0$, there exists a positive integer N such that $|a_n| < \varepsilon$ for all $n \geq N$.

Solution. We prove the result in a few steps.

Claim 1. $|a_{n+3}| \leq \max\{|a_n|, |a_{n+1}|\}$ for any $n \geq 1$.

Proof. WLOG assume $a_{n+3} > 0$ (if $a_{n+3} < 0$, replace a_i by $-a_i$ and replace $f(x)$ by $-f(-x)$). This implies $f(a_{n+2}) > f(a_{n+1})$, and hence $a_{n+2} > a_{n+1}$. This implies

$$a_{n+3} = f(a_{n+2}) - f(a_{n+1}) < a_{n+2} - a_{n+1}.$$

- If $a_{n+2} \leq 0$, then $a_{n+3} \leq -a_{n+1} \leq |a_{n+1}|$.

- If $a_{n+2} > 0$, then $f(a_{n+1}) > f(a_n)$, and hence $a_{n+1} > a_n$. This implies

$$a_{n+2} = f(a_{n+1}) - f(a_n) < a_{n+1} - a_n.$$

Therefore,

$$a_{n+3} < a_{n+2} - a_{n+1} < -a_n \leq |a_n|.$$

In either case, claim 1 is true. $\square$

Let $b_n = \max\{|a_n|, |a_{n+1}|, |a_{n+2}|\}$. By claim 1, we have $b_{n+1} \leq b_n$ for all n . As $\{b_n\}$ is bounded below by 0, it converges to some limit L by the monotone convergence theorem. If $L = 0$, then $\{a_n\}$ also converges to 0, which is our final goal. Therefore, we may now assume $L > 0$. WLOG assume $b_n \in [L, L+1]$ for all n by ignoring the initial terms.

Claim 2. There exists $c > 0$ such that for any x, y satisfying $|x|, |y| \leq L+1$ and $y - x \geq \frac{L}{2}$, we have

$$f(y) - f(x) \leq y - x - c.$$

Proof. Firstly, f is continuous since it satisfies the Lipschitz condition (1). (For any $\varepsilon > 0$, let $\delta = \varepsilon > 0$. Then for any $y > x$ with $y - x < \delta$, we have $0 \leq f(y) - f(x) < y - x < \delta = \varepsilon$.) Therefore, the function

$$g(x) = f(x) - f\left(x + \frac{L}{2}\right) + \frac{L}{2}$$

is continuous. If we restrict the domain of g to the closed interval $\left[-L-1, \frac{L}{2}+1\right]$, then g attains a minimum value c . Now, for any x, y satisfying $|x|, |y| \leq L+1$ and $y - x \geq \frac{L}{2}$, we must have $x \in \left[-L-1, \frac{L}{2}+1\right]$. Therefore, we have

$$f(x) - f\left(x + \frac{L}{2}\right) + \frac{L}{2} \geq c. \quad (2)$$

Since $y \geq x + \frac{L}{2}$, we have

$$f(y) - f\left(x + \frac{L}{2}\right) < y - \left(x + \frac{L}{2}\right). \quad (3)$$

Combining (2) and (3), we obtain

$$f(y) - f(x) < y - \left(x + \frac{L}{2}\right) - \left(c - \frac{L}{2}\right) = y - x - c$$

as desired. $\square$

Claim 3. $|a_{n+3}| \leq \max\left\{|a_n| - c, |a_{n+1}| - c, \frac{L}{2}\right\}$ for any $n \geq 1$.

Proof. The proof is similar to that of claim 1. Firstly, we may assume $a_{n+3} > 0$. The result is obvious if $a_{n+3} \leq \frac{L}{2}$, so we may assume $a_{n+3} > \frac{L}{2}$. As before, we have

$$a_{n+3} < a_{n+2} - a_{n+1}.$$

This implies $a_{n+2} - a_{n+1} > \frac{L}{2}$. By claim 2, we obtain

$$a_{n+3} = f(a_{n+2}) - f(a_{n+1}) \leq a_{n+2} - a_{n+1} - c.$$

- If $a_{n+2} \leq 0$, then $a_{n+3} \leq -a_{n+1} - c \leq |a_{n+1}| - c$.
- If $a_{n+2} > 0$, then $a_{n+1} > a_n$. This implies

$$a_{n+2} = f(a_{n+1}) - f(a_n) < a_{n+1} - a_n.$$

Therefore,

$$a_{n+3} \leq a_{n+2} - a_{n+1} - c < -a_n - c \leq |a_n| - c.$$

In either case, claim 3 is true. $\square$

Finally, let $\varepsilon > 0$ be a sufficiently small constant. As b_n converges to L , we have $b_n < L + c - \varepsilon$ for all large n . This implies $|a_n| < L + c - \varepsilon$ for all large n . By claim 3, we have

$$|a_{n+3}| \leq \max\left\{(L + c - \varepsilon) - c, (L + c - \varepsilon) - c, \frac{L}{2}\right\} = L - \varepsilon$$

for all large n . This contradicts the fact that $\{b_n\}$ converges to L . Thus, we are done. $\square$